

GHAYDA NA'IM JA'AFREH

Data Science & Artificial Intelligence Graduate | Python | Machine Learning | AI & Software Systems

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PROFESSIONAL SUMMARY

Data Science and Artificial Intelligence graduate with an Excellent GPA of 93.04%, ranked second in the Data Science and Artificial Intelligence graduating cohort. Hands-on experience across Python programming, machine learning, deep learning, computer vision, data analysis, backend integration, REST APIs, model evaluation, debugging, and technical documentation. Built applied projects involving healthcare AI, multimodal authentication, robotics integration, autonomous agents, image segmentation, interactive ML applications, and reproducible experimentation. Strong analytical and problem-solving skills with the ability to learn quickly, work independently or within teams, and translate technical requirements into practical solutions.

TECHNICAL SKILLS

Programming & Software: Python, SQL, C++, PHP, HTML, CSS, JavaScript, modular programming, debugging, input validation, error handling, Git, GitHub

AI & Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, machine learning, deep learning, classification, regression, CNNs, transfer learning, segmentation, explainable AI

Data & Analytics: Pandas, NumPy, data cleaning, preprocessing, feature engineering, visualization, model evaluation, comparative experimentation

Backend & Integration: FastAPI, REST APIs, application integration, logging, dependency management, virtual environments, interface validation

Computer Vision: OpenCV, image processing, instance segmentation, U-Net segmentation, OCR, feature extraction

Databases & Tools: SQLite, MySQL, Jupyter Notebook, Microsoft Excel, Tableau, Gradio, Pygame, Azure-hosted GPU environment

EDUCATION

Bachelor's Degree in Data Science and Artificial Intelligence | Mutah University | 2022 - 2026

GPA: 93.04% | Grade: Excellent | Ranked 2nd in the Data Science and Artificial Intelligence graduating cohort

SELECTED PROJECTS

StrokeLens - Secure & Explainable Healthcare AI Graduation Project

Graduation Project | Sep 2025 - Jun 2026 | Core Project Owner and Lead Technical Contributor

- Led development of a secure and explainable healthcare AI graduation project using structured clinical data and an integrated application workflow.
- Contributed across data preparation, AI experimentation, explainability, secure reporting, testing, system integration, and technical documentation.

Technologies: Python, Machine Learning, Explainable AI, Secure Systems

Privacy-Safe Multimodal Authentication System

Team Project | Main Model and Integration Contributor

- Integrated face verification, voice verification, speech recognition, multimodal fusion, and authentication logic into a privacy-aware Python/FastAPI workflow.
- Connected PyTorch, OpenCV, InsightFace, SpeechBrain, and Vosk components and resolved runtime, dependency, input-format, and integration issues through incremental testing and logging.

Technologies: Python, FastAPI, PyTorch, OpenCV, InsightFace, SpeechBrain, Vosk, REST API Integration

Tetris AI Lab - End-to-End Machine Learning Game Application

Individual Project | Public GitHub Release

- Developed and published a complete Python/Pygame Tetris AI application integrating a CNN policy with game logic, inference, safety controls, replay functionality, and multiple human/AI modes.
- Built a reproducible training-data pipeline with automated cleaning, video-level train/validation/test splitting, simulator-generated recovery states, and documented experiment tracking.

Technologies: Python, PyTorch, CNN, Pygame, data pipelines, testing, reproducible experimentation

Dr. NAO - AI-Assisted Dermatology Support System

Team Project | Main AI/Model and Robot-Integration Contributor

- Integrated AI model outputs with a NAO robot interaction workflow for a healthcare prototype and live demonstration.
- Supported system integration, robot interaction logic, testing, troubleshooting, demonstration preparation, and technical explanation.

Technologies: Python, AI, image analysis, NAO robot integration, system integration

Medical Image Segmentation & Computer Vision Portfolio

Kaggle and Academic Projects

- Built a teeth instance-segmentation pipeline for dental X-rays, achieving Mask mAP50 of 0.9947 and Mask mAP50-95 of 0.6301.
- Implemented a custom PyTorch U-Net workflow for dental-carries segmentation with augmentation, checkpointing, threshold selection, Dice, and IoU evaluation; achieved 0.9010 test Dice.
- Completed transfer learning for Intel image classification, MNIST CNN experiments, handwritten-letter OCR using HOG/KNN, image processing, and additional classification projects.

Technologies: Python, PyTorch, OpenCV, U-Net, transfer learning, OCR, segmentation

ML Decision Surfaces Lab - Interactive Machine Learning Playground

Team Project | Main Algorithm Developer and Interface Designer

- Built an interactive Gradio application for comparing classification and regression models using decision boundaries, ROC-AUC, confusion matrices, learning curves, and controlled noise/outlier experiments.

Technologies: Python, scikit-learn, Gradio, Matplotlib, model evaluation

EXPERIENCE

Data Science & Programming Trainee | The Hope International Company | Internship | Jun 20, 2025 - Sep 10, 2025

- Completed practical training in Python programming, data analysis, preprocessing, visualization, model implementation, debugging, teamwork, and technical documentation.
- Applied programming and analytical skills to structured datasets and small end-to-end AI workflows.

AI & Machine Learning Trainee | The Hope International Company | Internship | Nov 15, 2023 - Mar 15, 2024

- Built and evaluated Python-based machine-learning exercises covering feature selection, classification, regression, decision trees, KNN, Naive Bayes, and model evaluation.
- Developed an end-to-end understanding of the machine-learning lifecycle from data preparation through evaluation and reporting.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft | May 2026 | Credential ID: C5834AFC20D9A41C
- Data Science Essentials with Python | Cisco Networking Academy | May 2026 | Certificate ID: 960f2b70-a252-42a1-a73d-13e7ca5c7c51
- Python Essentials 1 | Cisco Networking Academy | May 2026 | Certificate ID: 01720f08-f6be-48a8-b730-d4da54339d1f
- Deep Learning - 60 Hours | The Hope International Company | Sep 2025 | Certificate ID: 1758145099980
- Python AI (Machine Learning) - 60 Hours | The Hope International Company | Mar 2024
- Ethical Hacker | Cisco Networking Academy | May 2026 | Certificate ID: 34898e2b-05cf-41a1-bcc0-c8d3f49090d1
- Cyber Security Associate | Green Circle for Software Solutions | Feb 2026
- CCNA Course Certificate | Pioneers Academy | Dec 2023 | Certificate No. 18437657-24
- Security+ Course Certificate - 30 Hours | Pioneers Academy | Oct 2023 | Certificate No. 18424325-2560
- Human Research: Data or Specimens Only Research - Basic Course | CITI Program, MIT Affiliates | Jul 2025 - Jul 2028 | Record ID: 70859362
- HTML Essentials | Cisco Networking Academy | May 2026 | Certificate ID: 1ca22c8d-bbae-4328-8cf8-07eacac17e5e
- International Computer Driving License (ICDL) - 36 Hours | Arab International Academy | Feb 2024
- English Communication Skills Training | Mutah University | May 2024
- English Conversation - 12 Levels, 75 Hours | Arab International Academy | Jun 2023 | Certificate No. 201548

LANGUAGES & PROFESSIONAL INTERESTS

Languages: Arabic (Native), English (Good)

Professional Interests: Artificial Intelligence, Machine Learning, Data Science, Computer Vision, Python Development, Backend Integration, Analytics, AI Research, and Software Engineering.